EXTENSION & ENRICHMENT WORKBOOK

SWIM & WATER *SAFETY*

Health & Physical Education · Year 7–8 · Victorian Curriculum F–10 Version 2.0

This workbook goes **deeper** on every topic. You will analyse real data, think critically about why drowning happens, evaluate evidence, and produce a sophisticated communication product. Expect to research, calculate, argue, and justify — not just describe.

STUDENT NAME _____

CLASS _____

TEACHER _____

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Extension Workbook — Expectations: Answers should go beyond description. *Analyse, evaluate, justify, and use evidence.* Where a question asks "why," give a reasoned explanation with supporting data, not just a one-word answer. Research questions require a source URL.

1 Statistical Analysis — Victorian Drowning Data

VC2HP8P10 · VC2HP8P09

ADVANCED CONTEXT

- → 52 fatal drownings in Victoria 2024–25 — below the 2020–21 peak of 64 and the 59 recorded in 2022–23, but 9% above the ten-year average of 48
- → Victoria's population ≈ 6.9 million → drowning rate = 52 ÷ 69 = 0.75 per 100,000
- → Australian national rate: 1.13 per 100,000 in 2025/26; ten-year average 1.12 (RLSSA National Drowning Report 2026). 313 deaths nationally, 10% above the ten-year average of 284. Coastal locations accounted for 151 deaths (48%), inland waterways 90 (29%) and swimming pools 41 (13%). 82% of those who drowned were male and 30% were aged 65 or over.
- → Non-fatal drownings: 60% above 10-year average — far more common than fatal
- → Multicultural communities: 37% of deaths since 2015–16 (185 deaths); country of birth unknown in 22% of cases
- → Males: 38 of 52 deaths (73%) — below the global pattern of ~80%, but still a large disparity
- → Zero deaths in public swimming pools (first time since 2019–20); 38% coastal, 29% inland, 25% in home environments
- → WHO: around 300,000 drowning deaths per year; 92% in low- and middle-income countries; global rate 3.8 per 100,000, down 38% since 2000

Sources: LSV — lsv.com.au/what-we-do/research-advocacy · RLSSA — royallifesaving.com.au · WHO — who.int/news-room/fact-sheets/detail/drowning

Activity 1.1 — Calculations & Interpretation

Individual · ~15 min

Question 1.1a — Calculation

2 marks

Using the population data above, calculate: *what percentage of the 52 Victorian drowning deaths were male?* Show your working.

Working: 38 ÷ 52 × 100 = _____%

Now compare this to the global figure (80%). What does the difference suggest about Victorian male risk-taking near water?

Victorian % is _____ than the global figure. This suggests _____

Question 1.1b — Rate Comparison 3 marks

Victoria's drowning rate was 0.75 per 100,000 in 2024–25. Australia's national rate was 1.13 per 100,000 in 2025/26. Calculate how many Victorian deaths would be expected if Victoria drowned at the national rate, then compare that with the 52 actually recorded. Show working, and interpret: what does the direction of the difference tell you, and what does it NOT tell you?

Actual deaths: 52 · National rate: 1.13 per 100,000 × 6.9 million = _____ expected deaths

Lives saved = 54 − _____ = _____

Interpretation:

Question 1.1c — CALD Data Analysis 3 marks

Life Saving Victoria reports 185 drowning deaths among people from multicultural backgrounds since 2015–16 — 37% of all Victorian drowning deaths — with country of birth unknown in a further 22% of cases. Using this information, explain whether multicultural Victorians are over-represented, under-represented, or proportionally represented in drowning deaths. Show your reasoning.

CALD % of drowning deaths: 40% · CALD % of population: 28%

This means CALD Victorians are: over-represented / under-represented / proportional (circle one)

Explanation:

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Suggest TWO specific reasons (not just "language barriers") why this over/under-representation exists:

1.

2.

Question 1.1d — Data Visualisation 3 marks

In the space below, create a **bar graph** comparing Victorian drowning deaths by location type. Research the approximate figures from the LSV report (lsv.com.au/what-we-do/research-advocacy). Label axes, include a descriptive title, and write a 2-sentence analysis beneath.

📊 *Draw your bar graph here — or attach on a separate page*

Analysis:

Source URL for your data: _____

2 Beach Flag System — Analysis & Critical Evaluation VC2HP8P10

Activity 2.1 — Flag System Evaluation ★ COLOUR

Individual · ~10 min

Beyond just identifying flags — analyse WHY each flag exists and WHO it protects. Complete the analysis column.



Red & Yellow

Why this flag exists + who it protects: _____



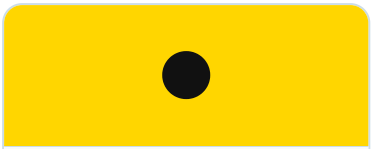
Single Red

Why this flag exists + limitation of this system: _____



Black & White

Why separation of zones matters + risk if ignored: _____



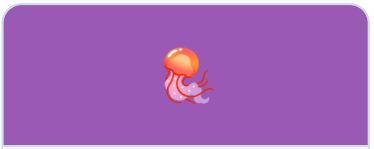
Yellow + Black Ball

Who enforces these rules + what happens if no-one does: ____



Orange Windsock

Physics reason: why offshore wind + inflatable = drowning risk: ____



Purple Flag

Limitation: why is this flag less reliable than the others? _____

Question 2.1a — Critical Evaluation 3 marks

The LSV data shows zero Victorian drowning deaths in public swimming pools in 2024–25, but 25% of deaths occurred in home environments. A student argues: "The flag system is pointless — people just ignore it anyway." Evaluate this argument. Do you agree or disagree? Use evidence.

I agree / disagree (circle) because: _____

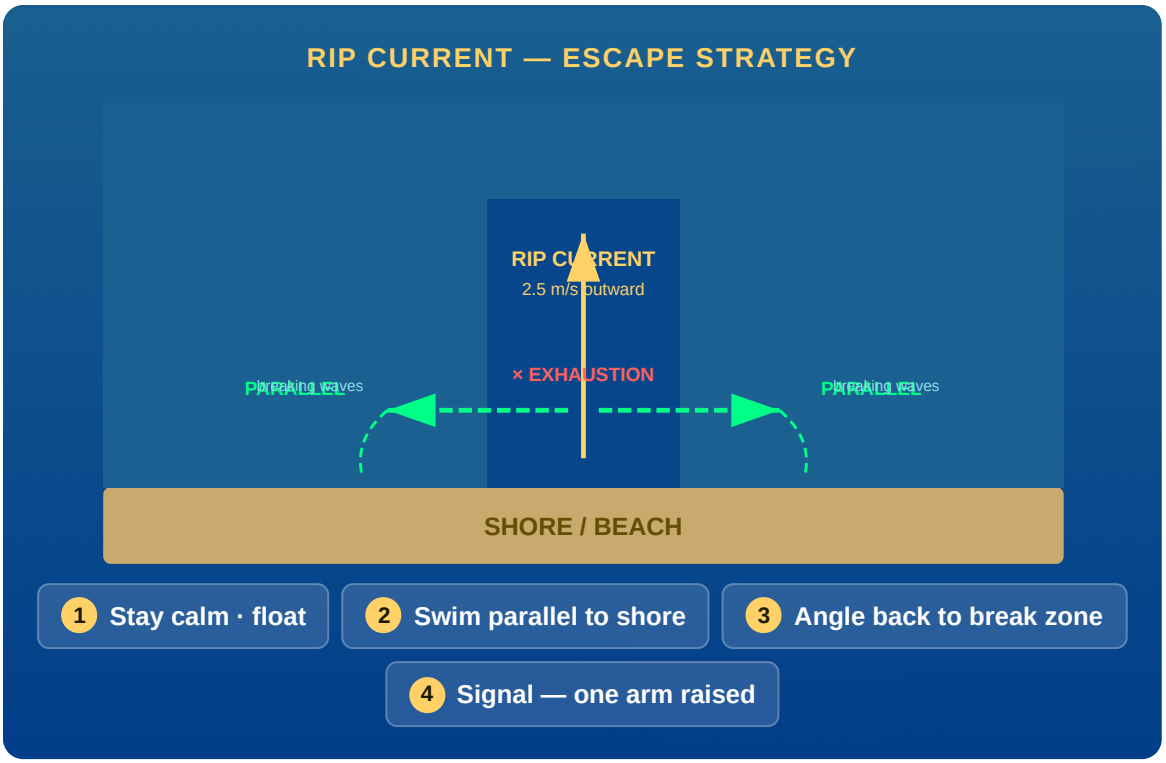
Evidence: _____

Counter-argument: _____

Activity 2.2 — Rip Current: Physics & Strategy ★ COLOUR

Individual · ~12 min

Rip currents can flow at up to 2.5 m/s. The world record 100m freestyle swim pace is approximately 2.0 m/s. Study the diagram and answer the analysis questions.



Question 2.2a — Physics Reasoning 3 marks

A rip current flows at 2.5 m/s. The world-record 100m swimmer averages approximately 2.0 m/s. Using this data, explain precisely why even a highly capable swimmer cannot escape a rip by swimming directly against it.

The rip current speed (2.5 m/s) is _____ than the world record swim speed (2.0 m/s).

This means that even the fastest swimmer in the world would move _____ m/s backward.

For an average Year 7 swimmer (~0.8 m/s), the net movement would be _____ m/s away from shore.

This leads to:

Question 2.2b — Strategy Evaluation 3 marks

The LSV advises floating and signalling for help rather than attempting to swim out of a rip if you are unable to escape in a few minutes. Evaluate this advice from a physiological standpoint — why does energy conservation matter more than attempting to escape?

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📊 **Research Extension**

Visit the SLSA rip current page (sls.com.au/surf-safety/rip-currents/) and find ONE statistic about rip currents that is NOT in your workbook. Write it below with the exact URL.

New statistic:

Source URL:

3 Emergency Response — Advanced Application

VC2HP8P10 · VC2HP8M01

DEEPER KNOWLEDGE

- → **CPR physiology:** Chest compressions generate ~25–30% of normal cardiac output — enough to keep the brain alive until the heart restarts
- → **4-minute rule:** Brain cells begin dying after 4–6 minutes without oxygenated blood — bystander CPR is the single most important survival factor
- → **Bystander effect:** Research shows bystander CPR rates are higher when individuals feel personally responsible — diffusion of responsibility in a crowd reduces the chance anyone acts
- → **Compression rate:** 100–120 beats per minute; depth 5–6cm; allow full chest recoil between compressions
- → **AED:** Automated External Defibrillators will only deliver a shock if the heart is in a shockable rhythm — they are safe for untrained bystanders to use

Source: St John Ambulance Vic — stjohnvic.com.au · Heart Foundation — heartfoundation.org.au

Activity 3.1 — Advanced Emergency Analysis

Individual · ~15 min

Question 3.1a — Physiological Explanation

3 marks

Explain the physiological reason why *starting CPR within 4 minutes* is more important than starting CPR after 8 minutes, even if professional help arrives at both times. Use the terms: cardiac output, oxygenated blood, brain cells, irreversible damage.

Question 3.1b — Bystander Effect

3 marks

The "bystander effect" describes how people in a crowd are less likely to help in an emergency because they assume someone else will act. Explain how you would specifically overcome the bystander effect at a beach incident to ensure CPR is started quickly.

Strategy: Instead of calling out to "someone," I would

Why this works psychologically:

EXTENDED SCENARIO

A group of 6 students are at an unpatrolled beach. One student is pulled under by a wave near rocks and is dragged back to shore unconscious after 3 minutes. Several bystanders are watching. You have a phone. The nearest lifeguard is 2km away at a patrolled beach. The nearest AED is at the car park, 400m away.

Question 3.1c — Multi-Step Decision Making

5 marks

Describe the exact sequence of actions for the first 5 minutes. Assign roles to the 6 people present, justify your decisions, and explain what you would do differently if the victim was found not breathing.

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Person	Assigned role	Why this role?
Person 1 (you)		
Person 2		
Person 3		
Persons 4–6		

If not breathing, the priority order changes because:

Question 3.1d — Throw Rescue Evaluation 3 marks

Rank and justify the throw rescue options at this scene. Available items: phone, rope attached to nearby boat, surfboard resting on beach, t-shirt, piece of driftwood. Explain the RLSSA hierarchy in your answer.

Item	Rank	Justification (with RLSSA principle)
Rope from boat		
Surfboard		
T-shirt		
Driftwood		
Phone		

4 Risk Analysis, Social Factors & Decision-Making

VC2HP8P08 · VC2HP8P10

Activity 4.1 — Social Determinants of Drowning Risk

Individual · ~15 min

Extension context: Drowning is not equally distributed across the population. Social factors — including gender norms, cultural background, and socioeconomic status — significantly influence who drowns and why. Research shows these factors often matter more than swimming ability alone.

Question 4.1a — Gender Analysis

3 marks

73% of Victorian drowning victims in 2024–25 were male. Research suggests this is partly explained by "masculinity norms" — social expectations about how males should behave. Explain what these norms are, how they increase drowning risk specifically, and propose ONE strategy a school could use to address this.

Question 4.1b — Hazard Mapping with Justification

4 marks

For each Victorian environment, identify THREE hazards, rank them H/M/L, and explain the mechanism by which each hazard causes drowning — not just that it's "dangerous."

Environment	Hazard	Rank	Mechanism (how it causes drowning)
River / Dam (e.g. Yarra, Eildon)			
Unpatrolled Beach (e.g. remote Victorian coast)			

Question 4.1c — Peer Pressure: Psychological Analysis

3 marks

Psychological research shows peer pressure is strongest when: (a) the group is watching, (b) there is a clear "dare" or challenge, and (c) refusing seems like weakness. Using this framework, design a TWO-STEP strategy for resisting a dare to swim at an unpatrolled location. Your strategy must address all three psychological triggers.

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5 Inland Waterways — Physiology, Research & Rescue

VC2HP8P10 · VC2HP8M01

Activity 5.1 — Cold Water Shock: Physiological Analysis

Individual · ~15 min

Question 5.1a — Physiological Sequence — 3 marks

Describe the complete physiological sequence that occurs when a person enters water at 12°C unexpectedly. Use the terms: gasping reflex, hyperventilation, peripheral vasoconstriction, cardiac stress, swimming failure. Explain why this is dangerous even in shallow water.

Question 5.1b — 1-10-1 Rule Application — 2 marks

Explain the 1-10-1 cold water survival rule and describe how you would apply it if you fell into Lake Eildon in December. What decisions would you make at each stage?

Minute 1 (breathing control):

Minutes 1–10 (swimming window):

Hour 1 (hypothermia risk):

Question 5.1c — Research Investigation — 3 marks

Visit Paddle Australia (paddle.org.au) and Parks Victoria (park.vic.gov.au). For ONE specific Victorian river or lake, research: (a) the specific hazards identified, (b) any seasonal factors (e.g. snowmelt, rainfall), and (c) any recent incidents if publicly available.

Waterway I researched:

Specific hazards:

Seasonal factors:

Relevant incident / safety note:

Source URLs:

Question 5.2 — Throw Rescue: Design Your Own Protocol 3 marks

You are kayaking on a remote Victorian river with a group of 4. One person's kayak overturns in cold water 15m from shore. They are conscious but struggling. Design a 3-step rescue protocol using the RLSSA hierarchy that accounts for the 15m distance, cold water, and remote location.

Reference: RLSSA — royallifesaving.com.au · Paddle Australia — paddle.org.au

Question 5.3 — Alcohol: Multi-Factor Risk Analysis 3 marks

Construct a multi-factor risk argument explaining why alcohol + cold Victorian river water is a particularly dangerous combination. Your answer must address at least FOUR distinct physiological or behavioural mechanisms and cite the relevant body system for each.

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Activity 5.2 — Diving, Spinal Injury & Structural Risk Analysis

Individual · ~15 min

DATA CONTEXT — AUSTRALIAN DIVING & SPINAL INJURY

- → Unsafe diving accounts for approximately **10%** of all spinal cord injuries and **20%** of quadriplegia cases in Australia (AIHW Diving Injury Report)
- → **95%** of aquatic spinal injury patients are male; **75%+** are under 30; **66%** suffer permanent disability
- → Highest incidence age group: **children aged 10–14**
- → The cervical vertebrae (C4–C6) fracture or dislocate on head-first impact — compressing or severing the spinal cord. Spinal cord neurons cannot regenerate; the damage is permanent.
- → Three 2024–25 case studies: **Terrigal Wharf NSW** (wharf over shallow tidal water), **Torquay Front Beach VIC** (shifting surf sandbank), **Bawley Beach NSW** (sandbank formed between visits — false confidence from prior experience)
- → Core prevention message: **"Feet First, First Time, Every Time"** — and check depth on every visit, not just the first. Sandbanks, tidal variation, drought, flooding, and sediment change depth without warning.

Sources: AIHW aihw.gov.au · Royal Perth Hospital spinal injury review · RLSSA royallifesaving.com.au

Question 5.2a — Structural Analysis

3 marks

The data shows 95% of aquatic spinal injury patients are male and the highest-risk age group is 10–14 years. Using the social determinants framework, construct a structural explanation for this pattern. Your answer must go beyond "males take more risks" — identify the specific social factors that drive the behaviour and explain why individual education alone is insufficient as a prevention strategy.

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Question 5.2b — False Confidence & Mechanism Analysis 3 marks

The Bawley Beach case (NSW 2024) illustrates *false confidence from prior experience*. Explain: (a) the specific physical mechanism by which a sandbank could form between visits and reduce water depth; (b) why prior safe experience at a location can be MORE dangerous than approaching it with no knowledge; and (c) what the "Feet First, First Time" message does and does not address as a prevention strategy.

Question 5.2c — Intersecting Risk Factors 4 marks

Using the data above and the social determinants content from Section 4, design a specific, evidence-based prevention strategy targeting the highest-risk group for aquatic spinal injury. Your strategy must: (a) name the target group precisely; (b) cite at least two statistics; (c) address both the physical hazard (depth change) and the social factor (masculinity norms/peer pressure); (d) include one SMART success metric; and (e) identify the most significant limitation of your approach.

6 Personal Action Plan & Communication Campaign

VC2HP8P09 · VC2HP8P10

🎯 Extension Assessment Task — Communication Campaign

Choose ONE option. All options require evidence-based argument, at least THREE credible sources with URLs, and a sophisticated understanding of your target audience.

Option A — Evidence-Based Social Campaign (targeted at a high-risk group)

Design a complete 3-slide social media campaign targeting ONE specific high-risk group identified in Victorian drowning data (e.g. young males, CALD communities, inland waterway users, alcohol-affected swimmers). Each slide must: address a specific barrier for that group, cite a statistic from LSV/WHO data, and include a measurable call to action. Justify your design choices in 100 words.

Option B — Policy Brief to a Government Minister

Write a 400–500 word policy brief to the Victorian Minister for Sport recommending TWO specific government actions to reduce the CALD drowning disproportionality. Your brief must: cite evidence, acknowledge counter-arguments, and propose a measurable success metric for each recommendation.

Option C — School Safety Audit Report ★

Conduct a water safety audit of your school: What does it currently provide? What are the gaps? Produce a structured 400-word report with specific recommendations benchmarked against Victorian Curriculum 2.0 requirements (VC2HP8P10, VC2HP8M01) and LSV classroom resources. Include a priority matrix.

Planning Space

Option chosen: _____ · Target audience / addressee:

_____

Central argument / key claim:

_____

Source 1 (URL):

_____

Source 2 (URL):

_____

Source 3 (URL):

_____

Evidence-based statistic I will use:

_____

✏️ Extension Self-Assessment

Rate yourself honestly — this is for your own reflection.

I can calculate and interpret drowning rate statistics (per 100,000)	Developing	Solid	Can teach it
I can explain the physics of rip currents numerically	Developing	Solid	Can teach it
I can explain CPR physiology and the 4-minute rule	Developing	Solid	Can teach it
I can analyse social determinants of drowning risk	Developing	Solid	Can teach it
I can explain the 1-10-1 cold water rule and apply it	Developing	Solid	Can teach it
I can produce an evidence-based policy argument	Developing	Solid	Can teach it

One thing I still want to investigate:

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`lsv.com.au · royallifesaving.com.au · who.int · victoriancurriculum.vcaa.vic.edu.au`

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